

Name			
USN			
Date	Conduction	Record	Viva
Allotted Marks			
Obtained Marks			

EXPERIMENT 6

Develop Pig Latin scripts to sort, group, join, project, and filter the data

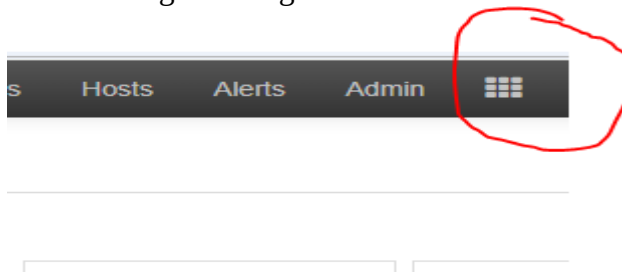
Aim: To develop a Pig Latin scripts to sort, group, join, and filter the data.

Theory:

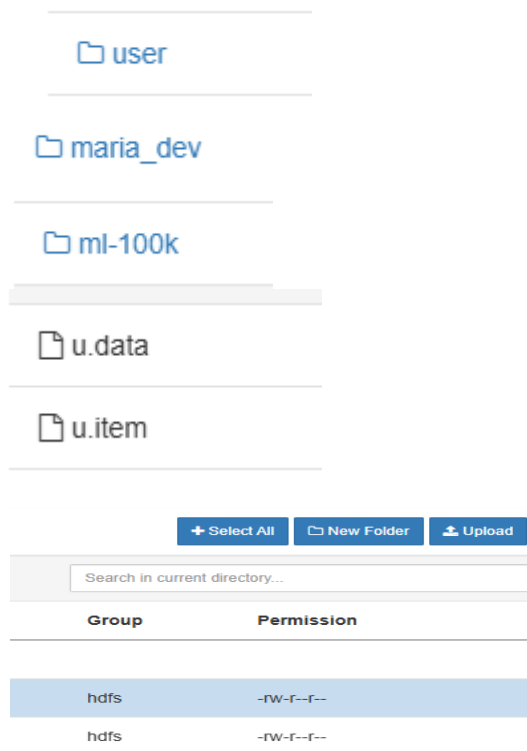
Pig Latin is a high-level scripting language used with **Apache Pig**, a platform for analyzing large data sets in **Big Data** environments. It provides an abstraction over **MapReduce**, making data processing simpler and more efficient. Pig Latin is a powerful tool in the Big Data ecosystem, especially for working with **Hadoop**. It simplifies data processing and analysis, making it a popular choice for handling large-scale data transformations. Writing MapReduce jobs in Java can be cumbersome. Pig Latin reduces the complexities. Can handle both **structured** and **semi-structured** data. Apache Pig automatically optimizes the execution of scripts. Works well with **Hadoop Distributed File System (HDFS)** and other Big Data tools. Allows users to define **User-Defined Functions (UDFs)** in Java, Python, or other languages.

Procedure:

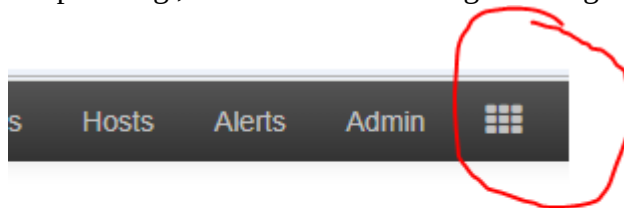
1. On the Ambari view go to the grid icon & select Files View



Navigate to user → maria_dev → ml-100k → u.data & u.item - upload



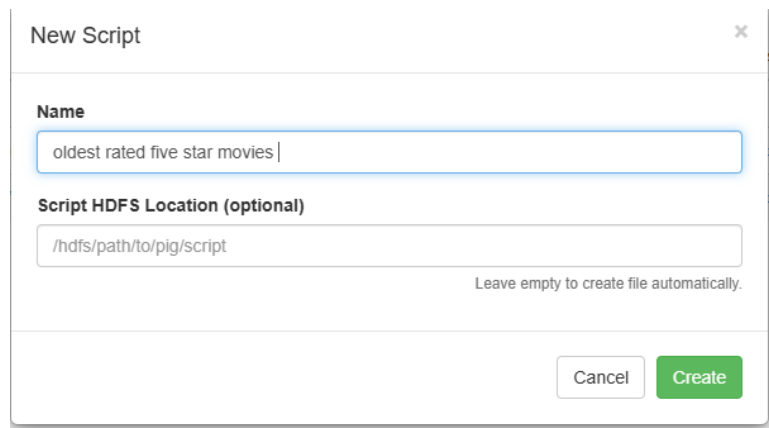
2. After uploading , On the Ambari view go to the grid icon & select Pig View



3. After selecting the Pig View , you will get a screen to start with Pig scripts, Go to New Script



After clicking New Script , you will get a screen , in the Name option type oldest rated five star movie and click create.



New Script

Name
oldest rated five star movies

Script HDFS Location (optional)
/hdfs/path/to/pig/script

Leave empty to create file automatically.

Cancel Create

4. After creating, type the following code

```
ratings = LOAD '/user/maria_dev/ml-100k/u.data' AS (userID:int, movieID:int, rating:
int, ratingTime:int);

metadata = LOAD '/user/maria_dev/ml-100k/u.item' USING PigStorage('|')
AS (movieID: int, movieTitle:chararray, releaseDate:chararray, videoRelease:chararray,
imdbLink: chararray);

nameLookup = FOREACH metadata GENERATE movieID, movieTitle,
ToUnixTime(ToDate(releaseDate, 'dd-MMM-yyyy')) AS releaseTime;

ratingsByMovie = GROUP ratings BY movieID;

avgRatings = FOREACH ratingsByMovie GENERATE group AS movieID,
AVG(ratings.rating) AS avgRating;

fiveStarMovies = FILTER avgRatings BY avgRating > 4.0;

fiveStarsWithData = JOIN fiveStarMovies BY movieID, nameLookup BY movieID;

oldestFiveStarMovies = ORDER fiveStarsWithData BY nameLookup::releaseTime;

DUMP oldestFiveStarMovies;
```

Then execute

oldest rated five star movies 

☐ Execute on Tez

PIG helper  UDF helper  `/user/admin/pig/scripts/oldest Rated five star movies-2025-03-14_05-07.pig`

```

1 ratings = LOAD '/user/maria_dev/ml-100k/u.data' AS (userID:int, movieID:int, rating: int, ratingTime:int);
2
3 metadata = LOAD '/user/maria_dev/ml-100k/u.item' USING PigStorage(',')
4 AS (movieID: int, movieTitle:chararray, releaseDate:chararray, videoRelease:chararray, imdbLink: chararray);
5
6 nameLookup = FOREACH metadata GENERATE movieID, movieTitle,
7 ToUnixTime(ToDate(releaseDate, 'dd-MMM-yyyy')) AS releaseTime;
8
9 ratingsByMovie = GROUP ratings BY movieID;
10
11 avgRatings = FOREACH ratingsByMovie GENERATE group AS movieID, AVG(ratings.rating) AS avgRating;
12
13 fiveStarMovies = FILTER avgRatings BY avgRating > 4.0;
14
15 fiveStarsWithData = JOIN fiveStarMovies BY movieID, nameLookup BY movieID;
16
17 oldestFiveStarMovies = ORDER fiveStarsWithData BY nameLookup::releaseTime;
18
19 DUMP oldestFiveStarMovies;

```

5. Check the Result

oldest rated five star movies - **COMPLETED**

Job ID job_1743135539171_0010

Started 2025-03-28 12:49

▼ Results

```

(493,4.15,493,Thin Man, The (1934),-1136073600)
(604,4.012345679012346,604,It Happened One Night (1934),-1136073600)
(615,4.0508474576271185,615,39 Steps, The (1935),-1104537600)
(1203,4.0476190476190474,1203,Top Hat (1935),-1104537600)
(613,4.037037037037037,613,My Man Godfrey (1936),-1073001600)
(633,4.057971014492754,633,Christmas Carol, A (1938),-1009843200)
(132,4.0772357723577235,132,Wizard of Oz, The (1939),-978307200)
(1122,5.0,1122,They Made Me a Criminal (1939),-978307200)
(136,4.123809523809523,136,Mr. Smith Goes to Washington (1939),-978307200)
(478,4.115384615384615,478,Philadelphia Story, The (1940),-946771200)
(524,4.021739130434782,524,Great Dictator, The (1940),-946771200)
(484,4.2101449275362315,484,Maltese Falcon, The (1941),-915148800)
(134,4.292929292929293,134,Citizen Kane (1941),-915148800)
(483,4.45679012345679,483,Casablanca (1942),-883612800)
(659,4.078260869565217,659,Arsenic and Old Lace (1944),-820540800)
(611,4.1,611,Laura (1944),-820540800)
(496,4.121212121212121,496,It's a Wonderful Life (1946),-757382400)
(525,4.027397260273973,525,Big Sleep, The (1946),-757382400)
(489,4.115384615384615,489,Notorious (1946),-757382400)
(1064,4.25,1064,Crossfire (1947),-725846400)
(519,4.1,519,Treasure of the Sierra Madre, The (1948),-694310400)
(613,4.037037037037037,613,My Man Godfrey (1936),-1073001600)

```

6. Repeat the program for Least Rated movie

least rated movies

Execute on Tez
Execute

PIG helper
UDF helper
/user/admin/pig/scripts/least_rated_movies-2025-03-14_06-03.pig

```

1 ratings = LOAD '/user/maria_dev/ml-100k/u.data' AS (userID:int, movieID:int, rating: int, ratingTime:int);
2
3 metadata = LOAD '/user/maria_dev/ml-100k/u.item' USING PigStorage('|')
4 AS (movieID: int, movieTitle:chararray, releaseDate:chararray, videoRelease:chararray, imdbLink: chararray);
5
6 nameLookup = FOREACH metadata GENERATE movieID, movieTitle,
7 ToUnixTime(ToDate(releaseDate, 'dd-MMM-yyyy')) AS releaseTime;
8
9 ratingsByMovie = GROUP ratings BY movieID;
10
11 avgRatings = FOREACH ratingsByMovie GENERATE group AS movieID, AVG(ratings.rating) AS avgRating;
12
13 fiveStarMovies = FILTER avgRatings BY avgRating < 2.0;
14
15 fiveStarsWithData = JOIN fiveStarMovies BY movieID, nameLookup BY movieID;
16
17 oldestFiveStarMovies = ORDER fiveStarsWithData BY nameLookup::releaseTime;
18
19 DUMP oldestFiveStarMovies;

```

7. Check the Results

least rated movies - **COMPLETED**

Job ID job_1743135539171_0012

Started 2025-03-28 12:53

▼ Results

```

(1580,1.0,1580,Liebelei (1933),-1167609600)
(1566,1.0,1566,Man from Down Under, The (1943),-852076800)
(1584,1.0,1584,Symphonie pastorale, La (1946),-757382400)
(1582,1.0,1582,T-Men (1947),-725846400)
(1350,1.5,1350,Crows and Sparrows (1949),-662688000)
(1581,1.0,1581,Woman in Question, The (1950),-631152000)
(1577,1.0,1577,Death in the Garden (Mort en ce jardin, La) (1956),-441849600)
(1587,1.0,1587,Terror in a Texas Town (1958),-378691200)
(1373,1.0,1373,Good Morning (1971),34473600)
(1571,1.0,1571,Touki Bouki (Journey of the Hyena) (1973),94694400)
(1309,1.0,1309,Very Natural Thing, A (1974),126230400)
(1352,1.0,1352,Shadow of Angels (Schatten der Engel) (1976),189302400)
(1374,1.0,1374,Falling in Love Again (1980),315532800)
(1572,1.0,1572,Wend Kuuni (God's Gift) (1982),378691200)
(440,1.6428571428571428,440,Amityville II: The Possession (1982),378691200)
(453,1.9375,453,Jaws 3-D (1983),410227200)
(438,1.1666666666666667,438,Amityville 3-D (1983),410227200)

```
